Deligne–Lusztig theory

Programme

Talk 1. Structure of finite reductive groups - Niccolò Mecacci (11 November 2024)

Following [MT11] and [DM91]:

(i) Definition and examples of linear algebraic group ([MT11, Definition 1.1, Example 1.2, Example GL_n p.4-5]);

(ii) Linear algebraic group as closed subgroups of GL_n [MT11, Theorem 1.7];

(iii) Jordan decomposition of elements ([MT11, Proposition 2.2 and Theorem 2.5 (a-b)], then define unipotent groups [MT11, Definition 2.6]);

(iv) Connected components ([MT11, Definition 1.8 and Definition 1.11], recall that for linear algebraic groups irreducible=connected [MT11, Proposition 1.13], we will need $\mathbf{G}^\circ$ in the subsequent talks);

(v) Radical and unipotent radical, definition of reductive and semisimple groups ([MT11, Definition 6.13 and 6.14]);

(vi) Structure of connected reductive and semisimple groups ([MT11, Proposition 6.20 (a) and (c)], then [MT11, Corollary 8.22], then [MT11, Theorem 8.21 (d)]);

(vii) Classification of (semi-)simple linear algebraic groups ([MT11, Definition 9.10, Proposition 9.11, Theorem 9.13] for this need to define (maximal) tori, Weyl groups, and roots ([MT11, Definition 3.3, Definition 8.1])), classification of root data ([MT11, Theorem 9.6 and discussion following 9.13]);

(viii) Definition of Frobenius and Steinberg endomorphisms ([MT11, Example 21.1, define Frobenius associated with $\mathbb{F}_q$-structure, Example 21.2, Definition 21.3, Theorem 21.5]), then define *finite reductive group* as $\mathbf{G}^F$ with $\mathbf{G}$ a connected reductive linear algebraic group and F a Steinberg endomorphism;

(ix) Define duality ([GM20, Definition 1.5.17] and give examples: $\mathrm{GL}_n(q)^* = \mathrm{GL}_n(q)$, $\mathrm{SL}_n(q)^* = \mathrm{PGL}_n(q)$, $\mathrm{SO}_{2n+1}(q)^* = \mathrm{Sp}_{2n}(q)$...)

(x) Levi and parabolic subgroups (define Borel subgroups[MT11, Definition 6.3], recall that $\mathbf{B} = \mathbf{R}_u(\mathbf{B}) \rtimes \mathbf{T}$ and that $\mathbf{T} = \mathbf{C}_{\mathbf{G}}(\mathbf{T})$ [MT11, Theorem 4.4 and Proposition 8.11 (a)], now define Parabolic subgroups and Levi subgroups [DM91, Definition 1.8 and Definition 1.16], show that every parabolic subgroup has (non-unique) Levi decomposition [DM91, 1.17 and 1.18], finally show that Levi subgroups are exactly centralisers of tori [DM91, Proposition 1.21 and Proposition 1.22]: you need to recall that $\mathbf{Z}^\circ(\mathbf{G})$ is a torus by [MT11, Proposition 6.20 (a)]).

Talk 2. Deligne–Lusztig varieties and Lusztig functors - Edoardo Salati (18 November 2024)

Following [GM20, Section 3.3], [DM91, Chapter 11] and [CE04, Section 7.3 and 8.3]:

(i) Motivation: sell the idea of an inductive construction of $\mathrm{Irr}(\mathbf{G}^F)$;

(ii) Informal definition of ℓ-adic cohomology ([GM20, 2.2.1] and [DM91, Chapter 10 (up to Definition 10.3)]);

(iii) Definition of Deligne–Lusztig varieties ([CE04, Definition 7.6 (only $\mathbf{Y}_{\mathbf{V}}^{(\mathbf{G},F)}$)]);

(iv) Definition of Lusztig induction (define this on modules first [CE04, p.125] (see also [DM91, p.47]), then consider the associated (virtual) characters [DM91, Proposition 4.5], finally recover [GM20, Definition 3.3.2] and [DM91, Proposition 11.2] (notice that while $\mathbf{Y}_{\mathbf{V}}^{\mathbf{G},F}$ does not coincide with $\mathcal{L}^{-1}(\mathbf{V})$, the associated induction functors are equivalent as explained at the beginning of [CE04, p.126]));

(v) Definition of Lusztig restriction (a similar construction yields Lusztig restriction, start by considering the generalised restriction from [DM91, p.47] and obtain the definition on characters from [DM91, Proposition 11.2] and [GM20, Definition 3.3.2], finally observe that Lusztig induction and restriction are dual with respect to the usual scalar product on class function (see [GM20, Definition 3.3.2]));

(vi) Example: when we consider a Levi subgroup contained in an F-stable parabolic subgroup, this construction recovers the classical Harish-Chandra induction and restriction ([GM20, Proposition 3.3.3 and Definition 3.1.5] as well as [DM91, Example 4.6 (iii) and the comment following Definition 11.1]);

(vii) Mackey formula (define the Mackey formula as in [GM20, Theorem 7.2] and mention that this is expected to hold in general but only known in the case included in [GM20, Theorem 7.2]);

(viii) Independence of the parabolic subgroup (mention that this is always expected to be true and follows from the validity of the MacKey formula [GM20, Proposition 3.3.8]);

(ix) Transitivity [GM20, Theorem 3.3.6];

(x) Define the Green function [GM20, Definition 3.3.11];

(xi) Character formula [GM20, Theorem 3.3.12];

(xii) Consequences of the character formula (first state the degree formula [GM20, Proposition 3.3.18], prove [GM20, Corollary 3.3.19], finally using the latter result and the character formula compute the values of Lusztig induction on central elements).

Talk 3. Deligne–Lusztig characters (as special case of Lusztig induction) - Marco Albert (25 November 2024)

Following [GM20, Section 2.2] [DM91, Chapter 12-13]:

(i) Using Lusztig induction for $\mathbf{L} = \mathbf{T}$ a maximal torus, define Deligne–Lusztig characters [GM20, 2.2.6] (for consistency with previous talk denote $\mathbf{R}_{\mathbf{T},\mathbf{U}}^{\mathbf{G}}$ by $\mathbf{R}_{\mathbf{T}\leq\mathbf{B}}^{\mathbf{G}}$ as in [GM20, Definition 3.3.2]). Recall that $\mathbf{R}_{\mathbf{T}\leq\mathbf{B}}^{\mathbf{G}}$ is independent of the choice of the Borel subgroup $\mathbf{B}$ (see Talk 2 (viii));

(ii) Orthogonality up to conjugation: introduce the Scalar Product Formula [GM20, 2.2.8] and prove [GM20, Corollary 2.2.9 and 2.2.10];

(iii) Obtain the degree formula [GM20, Theorem 2.2.12] as a consequence of [GM20, Proposition 3.3.18] (see Talk 2 (xii));

(iv) Obtain the character formula [GM20, Theorem 2.2.16] as a consequence of [GM20, Theorem 3.3.12] (see Talk 2 (xi)). For this also notice that [GM20, Definition 3.3.11] for $\mathbf{L} = \mathbf{T}$ reduces

to [GM20, Definition 2.2.15];

(v) Define uniform class functions [GM20, Definition 2.2.25];

(vi) Prove [DM91, Proposition 12.12] using [GM20, Corollary 2.2.9] introduced in (iii) above;

(vii) Prove [DM91, Proposition 12.13]. For this you need to recall [GM20, Example 3.3.4];

(viii) Quick introduction to Alvis–Curtis duality: introduce [DM91, Definition 8.8 and Definition 9.1], state [GM20, Corollary 3.4.5] and [DM91, Corollary 9.4 and Proposition 12.8], finally state the following lemma: for every p-constant function f of $\mathbf{G}^F$ (see [GM20, p.243]) we have $D_{\mathbf{G}}(\chi f) = D_{\mathbf{G}}(\chi) f$ for all $\chi \in \mathrm{Irr}(\mathbf{G}^F)$ (this follows from [GM20, Proposition 3.3.16]);

(ix) Using the results recalled in (viii) above prove [DM91, Corollary 12.14];

(x) Using the above result prove [DM91, Proposition 13.1] (see also [GM20, Corollary 2.2.19]).

Talk 4. Rational Lusztig series - Gaëtan Mancini (2 December 2024)

Following [GM20, Section 2.4-2.5]:

(i) Introduce geometric conjugation and series [GM20, Lemma 2.3.1, Theorem 2.3.2, Definition 2.3.4];

(ii) Quickly discuss [GM20, 2.3.18, 2.3.19, 2.3.20];

(iii) Reformulation of geometric conjugation and series I [GM20, 2.4.22, 2.4.23, 2.4.24];

(iv) Introduce [GM20, 2.4.5 and Definition 2.4.6] and define indivisible pairs [GM20, 2.4.9-2.4.10];

(v) Reformulation of geometric conjugation and series II [GM20, Proposition 2.4.28 and Corollary 2.4.29];

(vi) Quickly introduce (Langlands) duality [GM20, Definition 1.5.17] and give a couple of examples (e.g $\mathrm{GL}_n^\star = \mathrm{GL}_n$, $\mathrm{SL}_n^\star = \mathrm{PGL}_n$, $\mathrm{SO}_{2n+1}^\star = \mathrm{Sp}_{2n}$);

(vii) State [GM20, Proposition 2.5.5 and Lemma 2.5.7] (without explicitly defining the map $(\lambda, n) \mapsto t_{\lambda,n}$);

(viii) Introduce [GM20, 2.5.12, Corollary 2.5.14, and Definition 2.5.17];

(ix) Prove [GM20, Example 2.5.19] and state [GM20, Theorem 2.5.24];

(x) Finally, obtain partition into rational Lusztig series [GM20, Definition 2.6.1, Theorem 2.6.2, and Remark 2.6.3].

Talk 5. Unipotent characters (as special case of Lusztig series) - Marie Roth (9-13 December 2024)

Following [GM20, Section 2.3 and Chapter 4]:

(i) Define unipotent characters as $\mathrm{Uch}(\mathbf{G}^F) = \mathcal{E}(\mathbf{G}^F, 1)$ and show that this is a geometric series (see remarks following [GM20, Definition 2.3.8 and Definition 2.6.1];

(ii) Example of unipotent characters $1_{\mathbf{G}}$ and $\mathrm{St}_{\mathbf{G}}$ [GM20, Example 2.3.23];

(iii) Discuss elementary characterisation [GM20, 2.3.10-2.3.12];

(iv) Classification of unipotent characters [GM20, Chapter 4];

(v) Generic behaviour of unipotent characters [GM20, Theorem 4.5.8];

(vi) Unipotent degrees of a spetsial reflection groups [BMM14].

Talk 6. Jordan decomposition of characters - (16 December 2024)

Following [GM20, Sections 2.6, 4.7]. Including:

(i) Jordan decomposition via Lusztig induction for $\mathbf{C}^{\circ}_{\mathbf{G}^*}(s)\mathbf{C}_{\mathbf{G}^*}(s)^{F^*} \leq \mathbf{L}^*$;

(ii) Jordan decomposition for $\mathbf{Z}(\mathbf{G}) = \mathbf{Z}^{\circ}(\mathbf{G})$;

(iii) Discuss disconnected centre;

(iv) Discuss commutativity with Lusztig induction [GM20, Section 4.7];

Talk 7. Generalised Harish-Chandra theory - Damiano Rossi (13 January 2025)

Following [GM20, Section 3.5, 4.6]. Including:

(i) Introduce polynomial order and E-split Levi subgroups and tori (see also [CE04, Chapter 13]);

(ii) Define e-HC series, e-cuspidal pairs, and order relation $\ll_e$;

(iii) Discuss the classical case $e = 1$ ([GM20, Section 3.1-3.2]);

(iv) Discuss the unipotent case ([GM20, 4.6.20-4.6.21]);

(v) Discuss maximal extendibility and its applications.

Talk 8. Blocks of finite reductive groups - Annika Bartelt (20 January 2025)

Following [CE04, Chapter 9], [Cab18]. Including:

(i) Defining characteristic case [Cab18, 1.18, 3.4] (see also [Rou24, Section 4.1]);

(ii) Non-defining characteristic: Broué–Michel's theorem, Hiss' theorem, define unipotent blocks [CE04, Chapter 9];

(iii) Broué's jordan decomposition for blocks [Cab18, 4.13-4.14]. This motivates the talk on Bonnafé–Dat–Rouquier.

Talk 9. Classification of (unipotent) blocks in non-defining characteristic - Damiano Rossi (27 January 2025)

Following [CE04, Chapter 22] and [CE99]. Including:

(i) Twisted induction for blocks and connected subpairs;

(ii) Classification of blocks in terms of ℓ'-cuspidal pairs;

(iii) Discuss the case of bad primes [KM15];

(iv) Description of non-unipotent characters in unipotent blocks;

(v) Discuss Brauer–Lusztig blocks and the distribution of characters in blocks in terms of e-HC series.

Talk 10. Jordan classification of blocks I - Caroline Lassueur (3 February 2025)

Following [Cab18, Section 9]. Including:

(i) Recall Broueé's result [Cab18, 4.14] and introduce the need for a Morita equivalence;

(ii) Introduce Morita equivalences;

(iii) Discuss Bonnafé–Rouquier;

(iv) Discuss Bonnafé–Dat–Rouquier.

Talk 11. Jordan classification of blocks II - Gunter Malle (10 February 2025)

Discuss Jordan decomposition for blocks with connected centre following [KM23].

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